Knotted light

Does topology alone hold a knot of light together, and what does a pulsing knot cost? Register entries H29 (step 1) and H30

kronspice simulation report · 2026-11-07 · 10 cells per L0, region ±6 L0, absorber 12 cells

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Summary

Rañada's knotted light (the hopfion) is an exact solution of Maxwell's equations in vacuum whose magnetic field lines are closed loops, every pair linked once. It was set in the kronspice FDTD field solver (*em/fdtd.go*) and run against the exact solution, to ask whether topology alone holds a knot of field energy together (H29, step 1). The incoming, contracting half of the same knot was then fed in through a Huygens surface, so the ledger books every joule that enters and leaves (H30), and the measured QED (Euler–Heisenberg) vacuum was tested as a possible source of that inward wave.

- **The topology is kept.** The exact field's traced B lines link once (+1.0000) at every time to t = 4 L0/c; the solver's magnetic helicity follows the exact value (0.9873 against 0.9824 at t = 4).
- **The energy is not held.** Energy within 2 L0 falls from 0.9659 (solver; exact 0.9662) at t = 0 to 0.0186 (exact 0.0200) by t = 3; the rms radius grows as $\sqrt{(1 + t^2)}$ (0.996 → 4.117 L0).
- **Every joule of the implosion and explosion is booked.** The surface puts in 0.9862 of a knot and takes out 0.9799; the difference, 6.6×10^{-3}, leaks numerically into the absorber. Ledger residual -1.8×10^{-13} of one knot's energy.
- **The knot survives the implosion.** At t = 0 two lines traced on the solver's own field link +1.0000 (gap 0.034 L0).
- **A pulsing knot costs one knot's energy per period from outside.** With period 8 L0/c it is rebuilt to 0.9571, 0.9579, 0.9573 (exact 0.9668, 0.9674, 0.9668), linking +1.0000 each time; the net work over the middle cycle is +0.0046 knot energies, equal to the leak (in = out).
- **The measured QED vacuum returns nothing.** Of an explosion, 6.1×10^{-28} comes back between the separated halves (round-off); at most about $6a \approx 1.25\times10^{-6}$ of the field energy could ever be redirected, against the 1 knot per period needed.

What is being tested

H29: particles as knots of field energy

The user hypothesis (2026-11-05) is that particles are knots of field energy, held together by their topology, since a knot cannot untie without being cut. Kelvin proposed the same for atoms in 1867 (vortex knots in the ether). Step 1 asks the question every ring model in the repository has met: under ordinary, linear Maxwell electrodynamics, does a knotted field stay together?

Standard prediction. Linking is conserved, because Maxwell's equations carry the topology, but the energy is not held: the knot expands at c while staying linked. Derrick's theorem (1964) says no static, finite lump of a linear field exists in three dimensions; holding a knot needs a nonlinear term (Skyrme 1961; Faddeev and Niemi 1997).

H30: implosion and explosion

The user hypothesis (2026-11-05, after Walter Russell's "rhythmic balanced interchange") is that a particle is a cycle of implosion and explosion, and that its energy books balance only when the inward half is counted with the outward one.

Standard prediction. Maxwell's equations are time-symmetric, so the contraction is the expansion run backwards. A knot rebuilt every period needs an incoming wave supplied from outside at one knot's energy per period, here a power U·c/(8 L0), where U is the knot's energy.

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The model

The field is the hopfion in the Bateman form of Kedia et al. (2013), with lengths in L0 and times in L0/c:

$$\alpha = (r^2 - t^2 - 1 + 2iz) / (r^2 - (t - i)^2)$$
$$\beta = 2(x - iy) / (r^2 - (t - i)^2)$$
$$F = E + icB = (\text{grad } \alpha) \times (\text{grad } \beta) = \text{curl}(\alpha \text{ grad } \beta)$$

- It is set in *em/fdtd.go* as discrete curls of Re(α grad β) and Im(α grad β), so div E = div B = 0 exactly on the grid. The helicity sign of the formula is checked numerically against Maxwell's equations (+1).
- The solver also carries the vector potential A in the temporal gauge (A ← A − Δt·E after each step), so curl A = B at every step and the magnetic helicity $\int A \cdot B \, dV$, the integral measure of field-line linking, can be followed. For a null field such as the hopfion (E·B = 0, |E| = c|B|) it is conserved exactly.
- Linking of two field lines is the Gauss linking integral. Field-line seeds at (±0.6, 0, 0) L0.
- H30: a closed Huygens surface (total-field/scattered-field; Taflove and Hagness ch. 5), a cube of half-width 5.75 L0, made of 160,080 edge currents and 158,700 magnetic face currents (*em.FaceSource*, with its own exact ledger term). The declared incoming wave stands in for whatever medium would have to supply it.
- One knot = 1.748×10^{-10} J at E0 = 1 V/m, L0 = 1 m.

Results

H29 step 1: the knot in the solver

t (L0/c)	within 2 L0, solver / exact	absorbed	rms radius (L0)	helicity, solver / exact	curl A − B	linking exact / solver (closed)
0.00	0.9659 / 0.9662	0.0000	0.996	1.0000 / 1.0000	0	+1.0000 / +1.0000 (yes)
0.50	0.9554 / 0.9564	0.0001	1.114	0.9999 / 0.9999	3.5×10^{-15}	+1.0000 / +1.0043 (yes)
1.00	0.9050 / 0.9102	0.0001	1.411	0.9992 / 0.9995	6.0×10^{-15}	+1.0000 / +1.0000 (yes)
1.50	0.7297 / 0.7513	0.0001	1.800	0.9982 / 0.9988	9.7×10^{-15}	+1.0000 / not closed
2.00	0.3990 / 0.3918	0.0001	2.233	0.9971 / 0.9976	1.3×10^{-14}	+1.0000 / not closed
3.00	0.0186 / 0.0200	0.0003	3.159	0.9935 / 0.9933	2.0×10^{-14}	+1.0000 / not closed
4.00	0.0012 / 0.0012	0.0007	4.117	0.9873 / 0.9824	2.9×10^{-14}	+1.0000 / not closed

Energies are fractions of the starting energy. Solver helicity at t = 0 is 0.9921 of the exact value (grid discretization at 10 cells per L0); the helicity columns are normalized to their own t = 0 values. Lines traced on the solver's field stop closing after t = 1 (grid errors turn near-closed loops into open ones), which is why helicity, an integral, is the solver's topology measure.

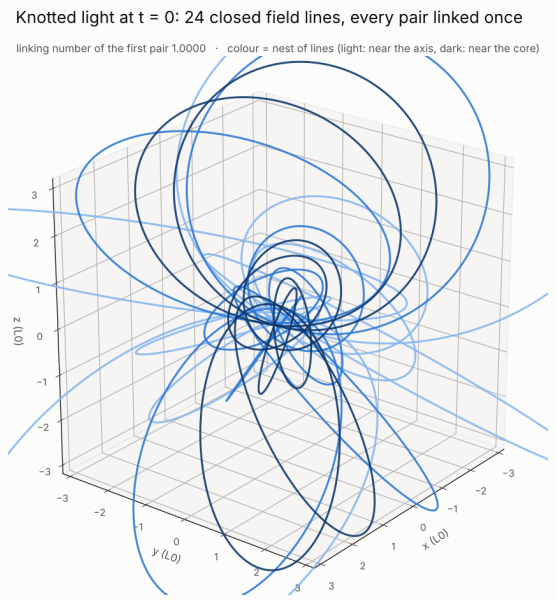


Figure 1. 24 magnetic field lines of the exact hopfion at t = 0, seeded on three nested rings (light: near the axis; dark: near the core). All close, and the first pair links once (1.0000).

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Figure 2. The same family of exact field lines at t = 0, 1, 2 and 3 L0/c on one scale, seeds scaled by the knot's size √(1 + t²). The grey sphere marks the energy's rms radius (1.00, 1.41, 2.23, 3.16 L0). All 24 lines close at every time and the linking stays 1.000, while the knot spreads at the speed of light.

- The slow fall of the helicity, in solver and exact solution alike, is the field's tail leaving the finite box, not the knot coming undone.
- Conclusion of step 1: linear Maxwell carries the knot's topology perfectly and does nothing to hold its energy. This is the standard prediction (Rañada 1989; Irvine and Bouwmeester 2008).

H30 Part 1: one implosion, then one explosion

t (L0/c)	within 2 L0, solver / exact	inside surface	put in / taken out	leak (numerical)	field error	linking (gap, L0)
−10	0.0000 / 0.0000	0.0041	0.0000 / 0.0000	0	0.000	—
−6	0.0000 / 0.0000	0.8198	0.8157 / 0.0000	7.2×10^{-8}	0.001	—
−4	0.0012 / 0.0012	0.9883	0.9842 / 0.0000	8.5×10^{-8}	0.003	—
−2	0.3773 / 0.3934	0.9902	0.9861 / 0.0000	8.5×10^{-8}	0.030	—
0	0.9567 / 0.9661	0.9903	0.9862 / 0.0000	8.5×10^{-8}	0.043	+1.0000 (0.034)
2	0.4129 / 0.3893	0.9902	0.9862 / 0.0001	8.8×10^{-8}	0.060	—
4	0.0012 / 0.0012	0.9883	0.9862 / 0.0020	4.5×10^{-7}	0.004	—
6	0.0000 / 0.0000	0.8390	0.9862 / 0.1498	1.6×10^{-3}	0.001	—
10	0.0000 / 0.0000	0.0038	0.9862 / 0.9799	6.6×10^{-3}	0.080	—

Energies as fractions of one knot's energy; selected rows of run_output.txt. Ledger residual -1.8×10^{-13} of one knot's energy.

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H30 Part 2: a pulsing knot

Knots tightest at t = 0, 8 and 16 L0/c are fed through the same surface; the first starts inside the surface at t = −4, the next two come in through it.

peak at t (L0/c)	within 2 L0, solver	exact	linking of two solver lines (gap)
0	0.9571	0.9668	+1.0000 (0.020 L0)
8	0.9579	0.9674	+1.0000 (0.035 L0)
16	0.9573	0.9668	+1.0000 (0.037 L0)

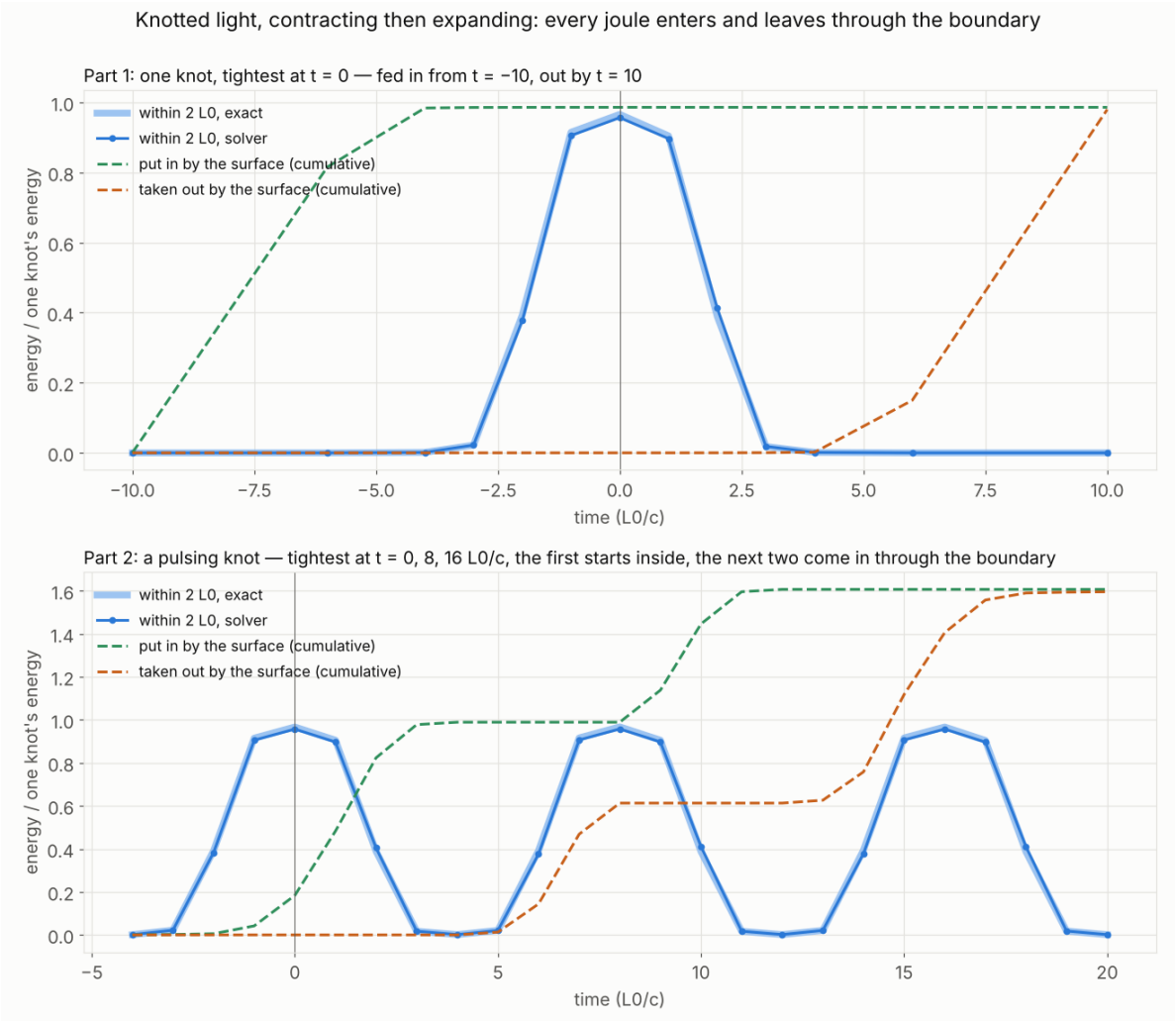


Figure 3. Top: Part 1, one knot fed in from t = −10 and out by t = 10. Bottom: Part 2, the pulsing knot. Blue: energy within 2 L0 (thick light line exact, dots solver). Dashed green and orange: the surface's cumulative work in and out. When an arriving and a departing knot overlap the columns show only the net flow, so the gross flow each way cannot be read from them.

- Over the middle cycle (t = 4 … 12) the surface's net work is +0.0046 knot energies, equal to the energy that leaked past it: in equals out. Ledger residual -3.7×10^{-13} of one knot's energy.
- Under linear Maxwell an imploding and exploding knot keeps its topology through both halves and can be kept pulsing indefinitely, at the cost of an inward wave of one knot's energy per period supplied from outside. The run books that inflow but supplies no source for it.

H30 Part 3: the measured QED vacuum as the source of the inward wave

Euler–Heisenberg's vacuum is the one nonlinear vacuum with experimental support (light-by-light scattering, ATLAS 2017, CMS 2019). Waves act on one another only where they cross; a lone plane wave never acts on itself. Both measurements run on Kron's plane-wave network with the Euler–Heisenberg vacuum as its nonlinear primitive (*kron.EHLine*, validated on light-by-light phase shifts to 0.4 %), beside Maxwell's linear vacuum, at 0.08–0.09 of the Schwinger field E_s (1.3×10^{18} V/m). Here $a = (\alpha/90\pi)(E/E_s)^2 = 2.09\times10^{-7}$ at 0.09 E_s.

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measurement	Euler–Heisenberg vacuum	needed for a knot rebuilt every period
A. Explosion into two halves: nonlinear energy back between the separated halves	6.1×10^{-28} of the explosion (linear lattice alone: 3.4×10^{-17})	1
Nonlinear part of the field after the halves crossed (phase shifts, all outgoing)	3.1×10^{-13}	—
B. Probe crossing a counter-propagating field of 0.08 E_s ($a = 1.65\times10^{-7}$): energy turned round	2.4×10^{-17} (Fresnel bound $(8a)^2 = 1.7\times10^{-12}$)	1
Nonlinear share of the field energy at the peak (the most the vacuum could redirect)	1.25×10^{-6} (≈ 6a)	1

- Nothing comes back: what the vacuum gives the crossing halves is a phase shift that travels outward with them; once apart, each is a plane wave the vacuum no longer touches.
- In three dimensions the expanding shell is locally a plane wave within a few L0, so the bound on what the vacuum could ever turn round is about 10^{-6} of the knot per period, against the whole knot needed.
- So this route is bounded. The Wheeler–Feynman absorber, the zero-point field and Born–Infeld remain untested.

Ledger and validation

check	result (run_output.txt)
H29 energy ledger residual	4.6×10^{-14} of the starting energy
max |div B| (relative)	5.2×10^{-15}
curl A − B	≤ 2.9×10^{-14} of max B (t = 4)
H30 Part 1 ledger residual	-1.8×10^{-13} of one knot's energy
H30 Part 2 ledger residual	-3.7×10^{-13} of one knot's energy

test	checks	held to
TestLinkingKnownCurves	Gauss linking integral: Hopf link = ±1, separated circles = 0	1e-3
TestHopfionExact	field is null (F·F = 0); traced B lines close and link once at t = 0 and t = 3	1e-12; 1e-3
TestKnotInSolver	coarse grid: energy spreads as exact; helicity kept; curl A = B; ledger	0.02; 0.02; 1e-12; 1e-10
TestKnotEnergy	energy density at t = 0 spherically symmetric; radial total = 3-D sum	1e-12; 2e-3
TestImplosionThroughSurface	≥ 0.9 of a knot enters; centre reaches exact energy; linking 1 at t = 0; in = out; ledger	0.05; 0.02; 0.05; 1e-10
TestVacuumInward	EH halves return nothing (below linear lattice ripple); reflection below Fresnel bound; nonlinear share 6a	1e-24; Fresnel; 1 %
em.TestFDTDMagneticSourceLedger	ledger closes with magnetic face sources; sources in the absorber refused	1e-10

Limits of the model

- The grid has 10 cells per L0; the solver's starting helicity is 0.9921 of the exact value, and after the wave has crossed 10 L0 of grid the solver's field is 4.3 % from the exact one within 2 L0.
- Field lines traced on the solver's field are fragile: they close only up to t = 1 in step 1, and in H30 each is closed at its nearest return (gaps 0.020–0.037 L0).
- The declared incoming wave in H30 stands in for a medium; the run gives no physical source for it.
- Part 3 is computed on a one-dimensional plane-wave network at the top of the Euler–Heisenberg weak-field range; the three-dimensional statement is an argument from the local plane-wave form of the expanding shell.

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Status

H29 step 1: bounded. Linear Maxwell keeps the topology and not the energy, which is the standard prediction; a holding knot needs a nonlinear term (step 2 is *examples/ether_knot*). **H30: bounded.** Linear Maxwell gives both halves by time symmetry; the pulsing knot needs an inward wave of one knot per period from outside; the measured QED vacuum is bounded as its source at about 10^{-6} or less of a knot per period; the absorber, zero-point field and Born–Infeld routes are untested.

Reproducing the results

```
go run ./examples/knotted_light               # step 1 and H30, about 3 min
go run ./examples/knotted_light -h29only      # step 1 only
go test ./examples/knotted_light              # -short skips the 3-D solve
go run ./examples/knotted_light -viz examples/knotted_light/viz   # traced 3-D lines
python3 examples/knotted_light/viz/render.py       # knot_3d.png, knot_time.png
python3 examples/knotted_light/viz/render_h30.py   # h30_cycle.png from run_output.txt
```

Sources: Rañada, Lett. Math. Phys. 18, 97 (1989); Irvine and Bouwmeester, Nature Physics 4, 716 (2008); Kedia et al., PRL 111, 150404 (2013); Taflove and Hagness, Computational Electrodynamics, 3rd ed., ch. 5; Wheeler and Feynman, Rev. Mod. Phys. 17, 157 (1945); Born and Infeld (1934); Heisenberg and Euler (1936); Moffatt, J. Fluid Mech. 35, 117 (1969); Derrick (1964); Skyrme (1961); Faddeev and Niemi, Nature 387, 58 (1997); Kelvin, Phil. Mag. 34, 15 (1867); Russell (1926, 1947).

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